

DATABASE DESIGN DOCUMENT: ALIGNED SURVEYORS WEB APP

1. Schema Architecture Overview

The database uses standard auto-incrementing integers for performance-critical lookup keys and sequential timeline streams. High-security boundaries and public-facing profile modules utilize universally unique identifiers (**UUIDs**) to block predictive enumeration. All entity relations are backed by foreign key integrity checks.

2. Model & Field Registry

2.1. User Model (User)

- **Domain:** Auth / Management Core
- **Description:** Handles team identities, administrative clearance permissions, and master account switches.
- **Fields:**
 - id: bigint (Primary Key, Auto-Increment)
 - name: string (Not Null)
 - email: string (Unique, Not Null)
 - email_verified_at: timestamp (Nullable)
 - password: string (Not Null)
 - is_admin: boolean (Default: false)
 - is_active: boolean (Default: true)
 - remember_token: string (Nullable)
 - created_at / updated_at: timestamps
- **Relationships:**
 - hasMany(ProjectUpdate::class, 'user_id')
 - hasMany(Message::class, 'user_id')
 - hasMany(Group::class, 'creator_id')
 - belongsToMany(Group::class, 'group_members', 'user_id', 'group_id')
 - belongsToMany(Group::class, 'group_admins', 'user_id', 'group_id')

2.2. Service Model (Service)

- **Domain:** Marketing / Capabilities Catalog
- **Description:** Lists fields of practice (e.g., Topographical, Mine, Engineering surveys) parsed read-only by the Gemini AI.
- **Fields:**
 - id: UUID (Primary Key)
 - title: string (Unique, Not Null)
 - slug: string (Unique, Not Null)
 - description: text (Not Null)

- created_at / updated_at: timestamps
- **Relationships:**
 - hasMany(Project::class, 'service_id')

2.3. Client Model (Client)

- **Domain:** Marketing / Corporate Portfolio
- **Description:** Tracks partners and firms Aligned Surveyors has contracted with (e.g., Mimosa, Zimplats, IDBZ).
- **Fields:**
 - id: UUID (Primary Key)
 - name: string (Unique, Not Null)
 - slug: string (Unique, Not Null)
 - is_visible: boolean (Default: true)
 - created_at / updated_at: timestamps
- **Relationships:**
 - hasMany(Project::class, 'client_id')

2.4. Project Model (Project)

- **Domain:** Shared Marketing / Internal Tracker
- **Description:** Represents specific engagements. Designed for low-text rendering overhead.
- **Fields:**
 - id: UUID (Primary Key)
 - service_id: UUID (Foreign Key $\rightarrow$ services.id, On Delete: RESTRICT)
 - client_id: UUID (Foreign Key $\rightarrow$ clients.id, On Delete: RESTRICT)
 - project_title: string (Not Null)
 - status: enum ('active', 'completed')
 - created_at / updated_at: timestamps
- **Relationships:**
 - belongsTo(Service::class, 'service_id')
 - belongsTo(Client::class, 'client_id')
 - hasMany(ProjectUpdate::class, 'project_id')

2.5. Project Update Model (ProjectUpdate)

- **Domain:** Internal Field Progress Logs
- **Description:** Sequential text-only log entries posted by engineers on location. Contains no files.
- **Fields:**
 - id: UUID (Primary Key)
 - project_id: UUID (Foreign Key $\rightarrow$ projects.id, On Delete: CASCADE)
 - user_id: bigint (Foreign Key $\rightarrow$ users.id, On Delete: RESTRICT)
 - update_text: text (Not Null)

- created_at / updated_at: timestamps
- **Relationships:**
 - belongsTo(Project::class, 'project_id')
 - belongsTo(User::class, 'user_id')

2.6. Asset Model (Asset)

- **Domain:** Marketing / Hardware Inventory
- **Description:** Standalone hardware registry (e.g., Total Stations, GNSS Receivers, UAVs) queried by Gemini to verify hardware competence.
- **Fields:**
 - id: UUID (Primary Key)
 - name: string (Not Null)
 - type: string (Not Null)
 - quantity: integer (Default: 1)
 - description: text (Nullable)
 - created_at / updated_at: timestamps
- **Relationships:** None

2.7. Conversation Model (Conversation)

- **Domain:** Chat / Peer-to-Peer
- **Description:** Direct messaging channel setup between exactly two internal system users.
- **Fields:**
 - id: UUID (Primary Key)
 - user_one_id: bigint (Foreign Key $\rightarrow$ users.id, On Delete: CASCADE)
 - user_two_id: bigint (Foreign Key $\rightarrow$ users.id, On Delete: CASCADE)
- **Database Indexes:** Composite unique index on ['user_one_id', 'user_two_id'] to prevent redundant direct lines.
- **Relationships:**
 - belongsTo(User::class, 'user_one_id')
 - belongsTo(User::class, 'user_two_id')
 - hasMany(Message::class, 'conversation_id')

2.8. Group Model (Group)

- **Domain:** Chat / Multi-User Channels
- **Description:** Multi-user chat rooms spawned by a team operator.
- **Fields:**
 - id: UUID (Primary Key)
 - name: string (Not Null)
 - creator_id: bigint (Foreign Key $\rightarrow$ users.id, On Delete: RESTRICT)
 - created_at / updated_at: timestamps
- **Relationships:**
 - belongsTo(User::class, 'creator_id')

- belongsToMany(User::class, 'group_members', 'group_id', 'user_id')
- belongsToMany(User::class, 'group_admins', 'group_id', 'user_id')
- hasMany(Message::class, 'group_id')

2.9. Group Member Model (GroupMember)

- **Domain:** Chat / Membership Pivot
- **Description:** Simple visibility table linking active users to specific chat rooms.
- **Fields:**
 - id: bigint (Primary Key, Auto-Increment)
 - group_id: UUID (Foreign Key $\rightarrow$ groups.id, On Delete: CASCADE)
 - user_id: bigint (Foreign Key $\rightarrow$ users.id, On Delete: CASCADE)
- **Database Indexes:** Unique composite key on ['group_id', 'user_id'].

2.10. Group Admin Model (GroupAdmin)

- **Domain:** Chat / Privilege Pivot
- **Description:** Tracks group-level moderators who possess permissions to evict members, add participants, and delete text records inside a specific room.
- **Fields:**
 - id: bigint (Primary Key, Auto-Increment)
 - group_id: UUID (Foreign Key $\rightarrow$ groups.id, On Delete: CASCADE)
 - user_id: bigint (Foreign Key $\rightarrow$ users.id, On Delete: CASCADE)
- **Database Indexes:** Unique composite key on ['group_id', 'user_id'].

2.11. Message Model (Message)

- **Domain:** Unified Chat Messaging Backend
- **Description:** Consolidated table storing textual communications for both peer channels and shared rooms.
- **Fields:**
 - id: bigint (Primary Key, Auto-Increment)
 - user_id: bigint (Foreign Key $\rightarrow$ users.id, On Delete: CASCADE)
 - conversation_id: UUID (Nullable, Foreign Key $\rightarrow$ conversations.id, On Delete: CASCADE)
 - group_id: UUID (Nullable, Foreign Key $\rightarrow$ groups.id, On Delete: CASCADE)
 - message_text: text (Not Null)
 - created_at: timestamp (Not Null)
- **Relationships:**
 - belongsTo(User::class, 'user_id')
 - belongsTo(Conversation::class, 'conversation_id')
 - belongsTo(Group::class, 'group_id')

3. Core Database Layer Integrity Rules

3.1. Destination Constraints

To guarantee that a single row in the messages table maps exclusively to either a private conversation or a group chat room, the database layer enforces an explicit Check Constraint:

```
SQL
ALTER TABLE messages
ADD CONSTRAINT chk_exclusive_chat_destination
CHECK (
    (conversation_id IS NOT NULL AND group_id IS NULL) OR
    (group_id IS NOT NULL AND conversation_id IS NULL)
);
```

3.2. Automated Creator Initialization Transaction

When a new instance is declared inside the groups table, the system must execute the creation logic sequentially inside a single database transaction block. This forces the initializing user's ID into both the visibility and admin pivot arrays at the database level:

```
PHP
DB::transaction(function () use ($request) {
    $group = Group::create([
        'name' => $request->name,
        'creator_id' => $request->user()->id,
    ]);

    // Attach creator to group_members and group_admins pivot vectors
    $group->members()->attach($request->user()->id);
    $group->admins()->attach($request->user()->id);
});
```